Traceable Retrieval and Citation Engine (TRACE)

Toward a high-trust, locally deployed scientific agent

Thienan Hoang and Austin R.J. Downey and Kerry Sado

UNIVERSITY OF
South Carolina
Molinaroli College of
Engineering and Computing

A High-Trust Scientific Agent

- An agent that reads our papers, reports, and data, then answers with sources
- High trust = traceable and repeatable
  - Same question, same cited answer, today and decades from now
- We build it ourselves so we know what every step does
  - Parser, database, model, and tools are all inspectable

Cites its sources

Every claim links to the document, page, and verbatim text

Parses science faithfully

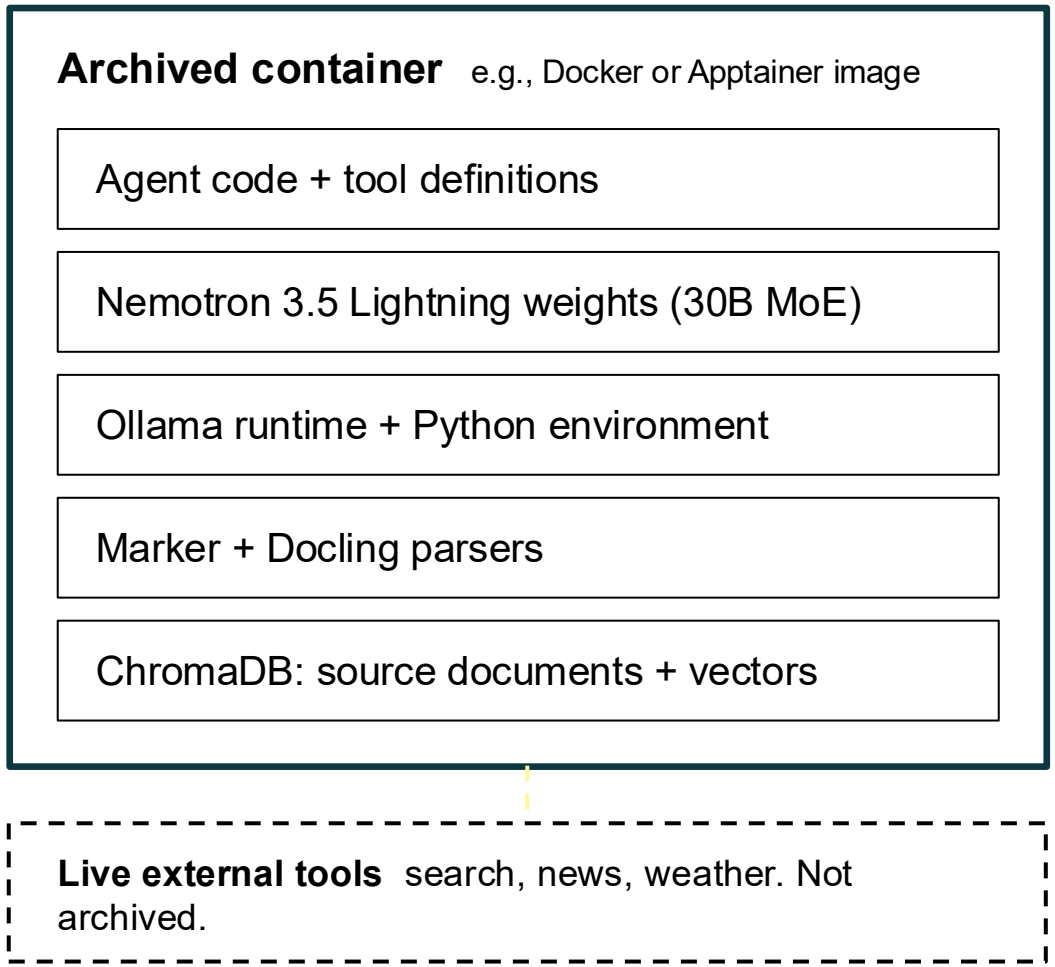
Tables, figures, equations, and references survive ingestion

Frozen and local

Same weights, same answer. No vendor can change the model on us

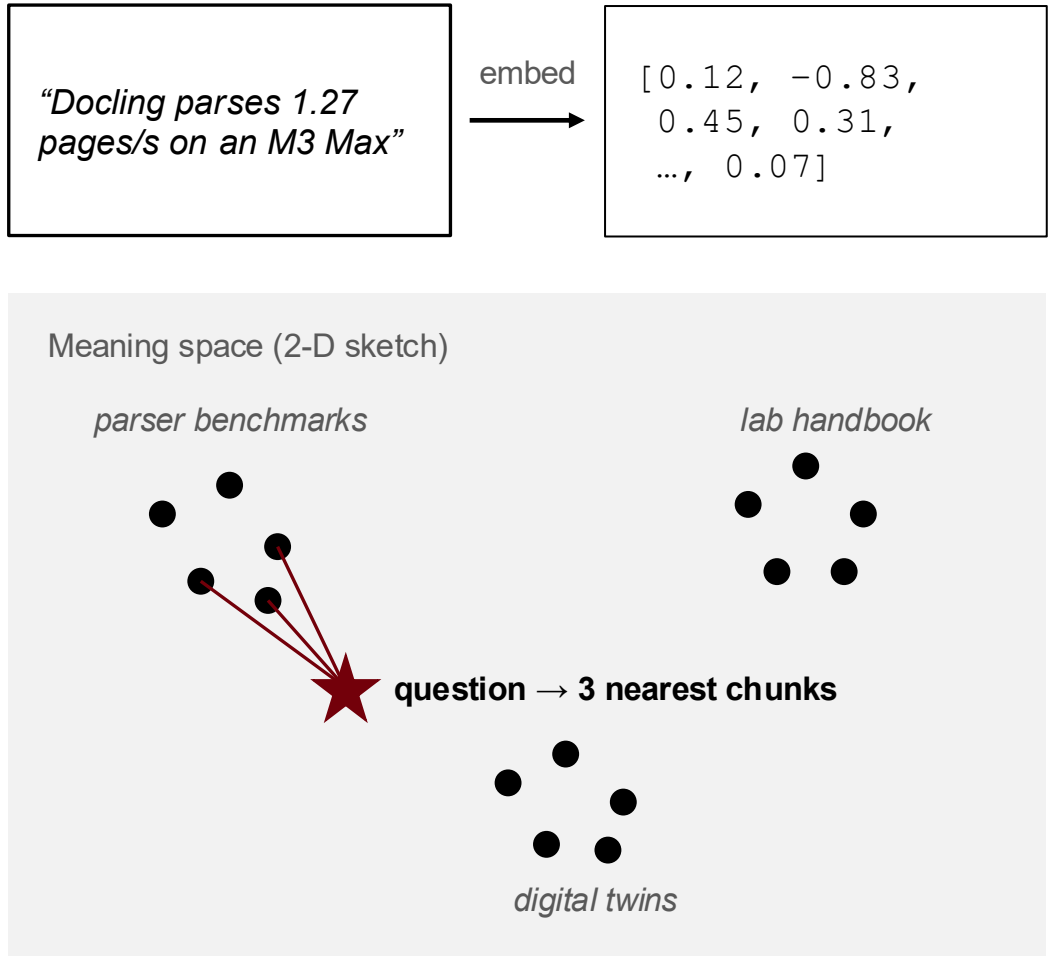
Packaged to Run for Decades

- One container holds everything
  - Model weights, runtime, parsers, database, agent code
- Pinned versions, fixed seed, zero temperature
  - Same query, same answer
- Archive it with the project record
  - Reopen and rerun in 10 to 20 years
- Live web tools stay outside the archive



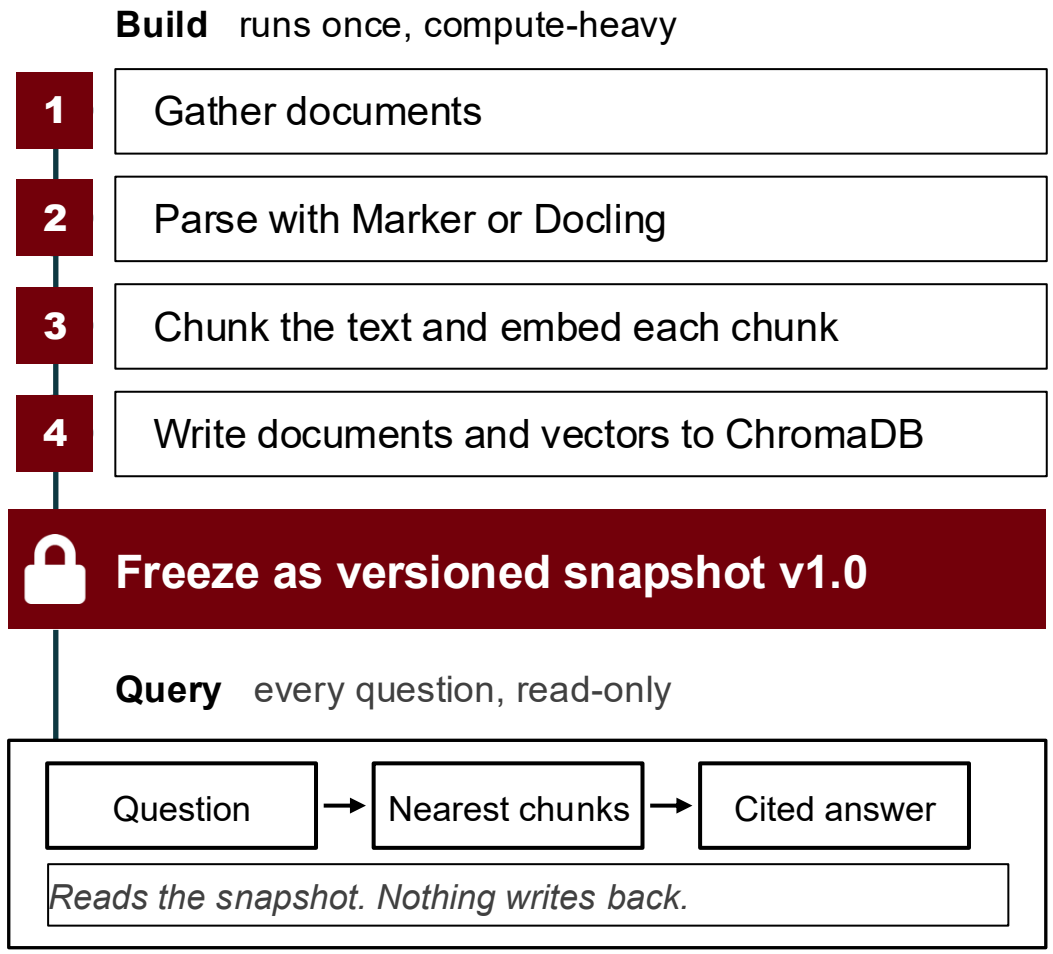
Vector Databases in One Slide

- Documents are split into short text chunks
- An embedding model turns each chunk into a vector
  - A list of hundreds of numbers that encodes meaning
- Similar meaning lands close together
- A question is embedded the same way and matched to its nearest chunks



Build Once, Then Freeze

- Build runs once
  - Gather, parse, chunk, embed, write
- Freeze the database, read-only from then on
- Query runs on every question
  - Reads the database, never writes
- The embedding model is frozen with it
- New documents mean a new versioned snapshot



RAG

- Document Classifier
  - To determine parser and profile to use for optimal system resource usage and time.
- Marker and Docling
  - Parsing engines
  - Can parse multiple formats
    - PDF, DOCX, CSV, PPTX, XLSX
  - Both capable of OCR
    - Docling: EasyOCR
    - Marker: Surya
  - Marker for scientific papers and Docling for general documents.



- https://github.com/datalab-to/marker.git
- https://github.com/docling-project/docling.git

Parsing Docling

- Current capabilities
  - Pulls text, images, and tables.
  - Identification of object type via optical character recognition.
    - Table, images, captions, body text, Formulas etc.
- Example document: Docling Technical Report



arXiv:2408.09869v5 [cs.CL] 9 Dec 2024

Docling Technical Report

Version 1.0

Christoph Auer Maksym Lysak Ahmed Nassar Michele Dolfi Nikolaos Livathinos
Panos Vagenas Cesar Berrospi Ramis Matteo Omenetti Fabian Lindlbauer
Kasper Dinkla Lokesh Mishra Yusik Kim Shubham Gupta Rafael Teixeira de Lima
Valery Weber Lucas Morin Ingmar Meijer Viktor Kuropiatnyk Peter W. J. Staar

AI4K Group, IBM Research
Rüschlikon, Switzerland

Abstract

This technical report introduces *Docling*, an easy to use, self-contained, MIT-licensed open-source package for PDF document conversion. It is powered by state-of-the-art specialized AI models for layout analysis (DocLayNet) and table structure recognition (TableFormer), and runs efficiently on commodity hardware in a small resource budget. The code interface allows for easy extensibility and addition of new features and models.

1 Introduction

Converting PDF documents back into a machine-processable format has been a major challenge for decades due to their huge variability in formats, weak standardization and printing-optimized characteristic, which discards most structural features and metadata. With the advent of LLMs and popular application patterns such as retrieval-augmented generation (RAG), leveraging the rich content embedded in PDFs has become ever more relevant. In the past decade, several powerful document understanding solutions have emerged on the market, most of which are commercial software, cloud offerings [3] and most recently, multi-modal vision-language models. As of today, only a handful of open-source tools cover PDF conversion, leaving a significant feature and quality gap to proprietary solutions.

With *Docling*, we open-source a very capable and efficient document conversion tool which builds on the powerful, specialized AI models and datasets for layout analysis and table structure recognition we developed and presented in the recent past [12, 13, 9]. Docling is designed as a simple, self-contained python library with permissive license, running entirely locally on commodity hardware. Its code architecture allows for easy extensibility and addition of new features and models.

Docling Technical Report 1

Auer, Christoph, et al. “Docling Technical Report.” *arXiv*, 2024, https://doi.org/10.48550/arXiv.2408.09869.

Docling Feature extraction (tables)

- Parsing of tables
  - Will preserve table header if table exceeds chunk size for maintaining semantic

torch runtimes backing the Docling pipeline. We will deliver updates on this topic at in a future version of this report.

Table 1: Runtime characteristics of Docling with the standard model pipeline and settings, on our test dataset of 225 pages, on two different systems. OCR is disabled. We show the time-to-solution (TTS), computed throughput in pages per second, and the peak memory used (resident set size) for both the Docling-native PDF backend and for the pypdfium backend, using 4 and 16 threads.

CPU	Thread budget	native backend			pypdfium backend		
		TTS	Pages/s	Mem	TTS	Pages/s	Mem
Apple M3 Max (16 cores)	4	177 s	1.27	6.20 GB	103 s	2.18	2.56 GB
	16	167 s	1.34		92 s	2.45	
Intel(R) Xeon E5-2690 (16 cores)	4	375 s	0.60	6.16 GB	239 s	0.94	2.42 GB
	16	244 s	0.92		143 s	1.57	

5 Applications

Thanks to the high-quality, richly structured document conversion achieved by Docling, its output qualifies for numerous downstream applications. For example, Docling can provide a base for detailed enterprise document search, passage retrieval or classification use-cases, or support knowledge extraction pipelines, allowing specific treatment of different structures in the document, such as tables, figures, section structure or references. For popular generative AI application patterns, such as retrieval-augmented generation (RAG), we provide *quackling*, an open-source package which capitalizes on Docling's feature-rich document output to enable document-native optimized vector embedding and chunking. It plugs in seamlessly with LLM frameworks such as LlamaIndex [8]. Since Docling is fast, stable and cheap to run, it also makes for an excellent choice to build document-derived datasets. With its powerful table structure recognition, it provides significant benefit to automated knowledge-base construction [11, 10]. Docling is also integrated within the open IBM data prep kit [6], which implements scalable data transforms to build large-scale multi-modal training datasets.

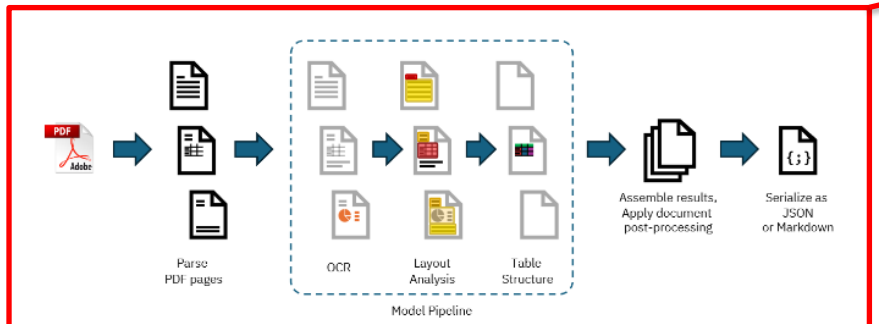
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(16 cores)	4	375 s 244 s	0.60 0.92	6.16 GB	239 s 143 s	0.94 1.57	2.42 GB

Docling Feature extraction (images)

- Image Extraction
  - Image object extracted via pipeline
  - Provides a uniform resource identifier (URI) to link image to parsed output.

Figure 1: Sketch of Docling's default processing pipeline. The inner part of the model pipeline is easily customizable and extensible.



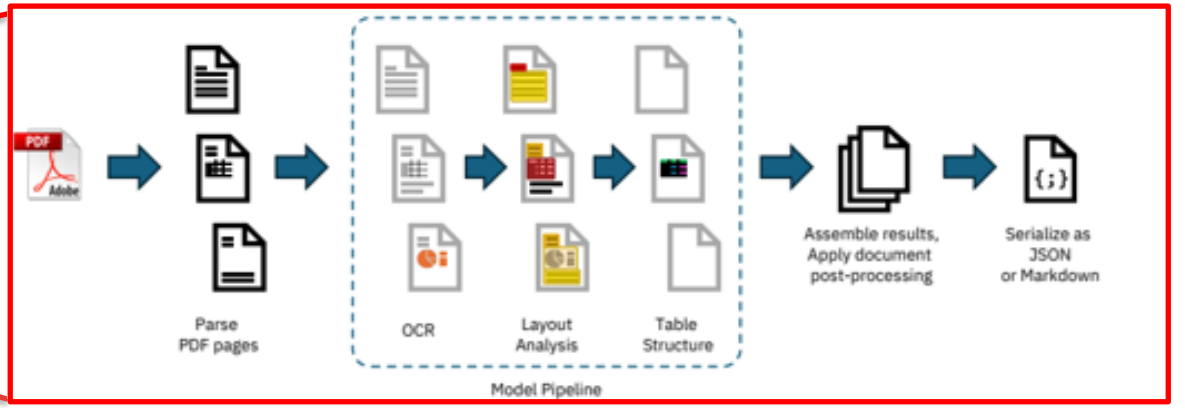


Figure 1: Sketch of Docling's default processing pipeline. The inner part of the model pipeline is easily customizable and extensible.

licensing (e.g. pymupdf [7]), poor speed or unrecoverable quality issues, such as merged text cells across far-apart text tokens or table columns (pypdfium, PyPDF) [15, 14].

We therefore decided to provide multiple backend choices, and additionally open-source a custom-built PDF parser, which is based on the low-level *qpdf* [4] library. It is made available in a separate package named *docling-parse* and powers the default PDF backend in Docling. As an alternative, we provide a PDF backend relying on *pypdfium*, which may be a safe backup choice in certain cases, e.g. if issues are seen with particular font encodings.

Docling Feature extraction (formulas)

- Formula extraction
  - Outputs in LaTeX format.

with $\mathbf{P} = (1-d_0)(1-d_1)e^{a_{11}\theta P+\left(-d+\frac{c_0\zeta_0\omega_3}{1+A_0\omega_3}\right)(1-\theta)P}$. The periodic solution $\widetilde{x}_4(t)$ is globally asymptotically stable from Lemma 4. Therefore, according to comparison theorem of impulsive differential equations [43], for any $\varepsilon > 0$, there exists a t_0 such that for $t > t_0$, $x_1(t) \leqslant x_4(t) \leqslant \widetilde{x}_4(t) + \varepsilon$. From system (2) again, we get

$$\begin{cases} \frac{dx_0}{dt} \geqslant x_0(t)(a_{10} - b_{10}\omega_3) & ,t \in (nP, (n+\theta)P) \\ \Delta x_0 \geqslant -\frac{h_{\max}}{\alpha_0}x_0 & ,t = (n+\theta)P \\ \frac{dx_0}{dt} \geqslant x_0\left(\frac{a_{20}}{1+\alpha(\widetilde{x}_4(t)+\varepsilon)} - b_{20}\omega_3 - \frac{\zeta_0(\widetilde{x}_4(t)+\varepsilon)}{1+B_0(\widetilde{x}_4(t)+\varepsilon)}\right) & ,t \in ((n+\theta)P, (n+1)P) \end{cases} \quad (7)$$

Then $x_0((n+1+P)P^+) \geqslant \rho_2 x_0((n+P)P^+)$, for all $t > n_1P$, where $n_1P \geqslant t_0$. Therefore, $x_0((n_1+k_1+\theta)P) \geqslant k_1\rho_2^{n_1}x_0((n_1+\theta)P^+) \to +\infty$ as $k_1 \to +\infty$, which implies $x_0(t) \to +\infty$ as $t \to +\infty$, which is contradiction. Thus there exists a $t_1 > 0$ such that $x_0(t_1) \geqslant \omega_3$.

If $x_0(t) \geqslant \omega_3$ for all $t \geqslant t_1$, then our objective is achieved. We therefore consider only those solutions whose trajectories leave and later re-enter the region defined by $R = \{x \in R_+^2 : x_0 < \omega_3\}$. Let $t^* = \inf_{t \geqslant t_1}\{x_0(t) < \omega_3\}$. Then $x_0(t) \geqslant \omega_3$ for $t \in [t_1, t^*)$.

Case A: with $t^* = (n_2+\theta)P$. Select natural numbers $n_3, n_4 \in \mathbb{N}$ such that


```
\begin{cases} \frac { d x _ { 0 } } { d t } \geqslant x _ { 0 } ( t ) ( a _ { 1 0 } - b _ { 1 0 } \omega _ { 3 } ) & , t \in ( n P , ( n + \theta ) P ) \\ \Delta x _ { 0 } \geqslant - \frac { h _ { m a x } } { a _ { 0 } } x _ { 0 } & , t = ( n + \theta ) P \\ \frac { d x _ { 0 } } { d t } \geqslant x _ { 0 } \left ( \frac { a _ { 2 0 } } { 1 + a ( \tilde { x } _ { 4 } ( t ) + \varepsilon ) } - b _ { 2 0 } \omega _ { 3 } - \frac { \zeta _ { 0 } ( \tilde { x } _ { 4 } ( t ) + \varepsilon ) } { 1 + B _ { 0 } ( \tilde { x } _ { 4 } ( t ) + \varepsilon ) } \right ) & , t \in ( ( n + \theta ) P , ( n + 1 ) P ) \end{cases}
```

$$\begin{cases} \frac{dx_0}{dt} \geqslant x_0(t)(a_{10} - b_{10}\omega_3) & ,t \in (nP, (n+\theta)P) \\ \Delta x_0 \geqslant -\frac{h_{max}}{a_0}x_0 & ,t = (n+\theta)P \\ \frac{dx_0}{dt} \geqslant x_0\left(\frac{a_{20}}{1+a(\widetilde{x}_4(t)+\varepsilon)} - b_{20}\omega_3 - \frac{\zeta_0(\widetilde{x}_4(t)+\varepsilon)}{1+B_0(\widetilde{x}_4(t)+\varepsilon)}\right) & ,t \in ((n+\theta)P, (n+1)P) \end{cases}$$

Gao, Bingying, et al. “Dynamics of a Fishery Management Model with Predation-Induced Fear Effect, Impulsive Nonlinear Harvesting Prey and Predator Seasonally Migrating Between Two Patches.” *Axioms*, vol. 15, no. 1, 2026, article 32, https://doi.org/10.3390/axioms15010032

Parsing Marker

- Largely same capabilities as Docling
  - More expensive, but better accuracy.
- Example document: A Digital Twin Based Forecasting Framework for Power Flow Management in DC Microgrids



A Digital Twin Based Forecasting Framework for Power Flow Management in DC Microgrids

Kerry Sado[1,*], Jarrett Peskar[2], Austin Downey[2,3], Jamil Khan[2,3], and Kristen Booth[1]

[1]University of South Carolina, Department of Electrical Engineering, Columbia, SC, USA
[2]University of South Carolina, Department of Mechanical Engineering, Columbia, SC, USA
[3]University of South Carolina, Department of Civil and Environmental Engineering, Columbia, SC, USA
*ksado@email.sc.edu
This work was supported by the Office of Naval Research under contracts N00014-22-C-1003 and N00014-23-C-1012. "The views expressed are those of the authors and do not reflect the official policy or position of the Department of Defense or the U.S. Government. DISTRIBUTION STATEMENT A. Approved for public release distribution unlimited. Approved. DCN# 2025-1-31-562"

ABSTRACT

The ability to forecast system conditions is integral to the definition and functionality of digital twins. While forecasting methods have been explored for use in digital twin systems, the integration of feedback mechanisms for real-time forecasting and in-situ decision-making in DC microgrids has not been extensively investigated. This research develops a modular forecasting framework tailored for digital twins in DC microgrids to enable real-time monitoring, online forecasting, and decision-making. DC microgrids, characterized by dynamic load variations, benefit from advanced predictive capabilities to maintain stability and operational efficiency. The proposed digital twin-based forecasting framework addresses these challenges by providing real-time predictive insights based on dynamic system conditions and a forecasting window defined by a decision-maker, facilitating proactive management strategies. Leveraging real-time sensor data, the digital twin forecasts system behavior under varying load conditions, enabling proactive management through real-time decision-making within operational constraints. As a proof of concept, the framework incorporates an electro-thermal digital twin designed to manage power flow based on thermal constraints in power distribution cables. Experimental validation using a simplified three-bus DC microgrid testbed demonstrates the effectiveness of the framework in enabling timely adjustments to power flows and preventing thermal overloads.

Introduction

A Digital Twin (DT) is a comprehensive digital replica of a physical asset that integrates multiphysics, various scales, and probabilities to mirror and forecast the life-cycle of its counterpart[1]. Essentially, the digital twin acts as a comprehensive and faithful representation of a physical system or subsystem, referred to as the physical twin[2]. In this context, a physical twin refers to an actual real-world entity which can vary in complexity. It might be a machine, a piece of infrastructure, a single system component, or an entire complex system. The physical twin encapsulates specific attributes, characteristics, and functionalities that define the entity. The definition of digital twins is often misused in the literature with many studies failing to emphasize the feedback loop that differentiates digital twins from traditional models. The bidirectional flow of data between the physical asset and its digital counterpart is integral to the digital twin paradigm as it provides real-time insights, updates, and enabling dynamic decision-making[3–5]. This mechanism enables the digital twin to adaptively update its state in real-time, driven by sensor data from the physical twin. Furthermore, the bidirectional nature of the feedback loop allows the digital twin to receive information about the current operational parameters from the physical twin, enabling synchronized adjustments in the virtual representation. The bidirectionality also allows the decision-making entity within the digital twin to execute instructions on the physical twin based on forecasted insights generated by the virtual representation. This capability ensures synchronized operations and dynamic adjustments. Without the bidirectionality, a model cannot be considered a true digital twin but rather a static digital representation. The omission of this crucial aspect in research underscores the potential benefits and applications of digital twin technology[3].

The significance of digital twins is rapidly gaining recognition from both academic and industrial sectors[6]. In the automotive industry, digital twins are used to monitor, simulate, and optimize production and operational performance[7]. In the energy sector, General Electric (GE) has implemented a digital twin for a wind farm, enhancing reliability by gathering real-time data on weather, performance, and service[8]. Similarly, Siemens developed a digital twin for the Finnish power grid, leading to improved safety, reliability, and resource savings[9]. Digital twin technology has applications beyond industrial and infrastructure sectors. In healthcare, digital twins are being explored to model cancer patients, providing patient-specific clinical decision

Sado, Kerry, et al. "A Digital Twin Based Forecasting Framework for Power Flow Management in DC Microgrids." *Scientific Reports*, vol. 15, 2025, article 6430, https://doi.org/10.1038/s41598-025-91074-0.

Marker Feature Extraction (in text citations)

- Linking in-text citations to the references section
  - Largely dependent on the document itself. (garbage in, garbage out)

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What Our Database Stores

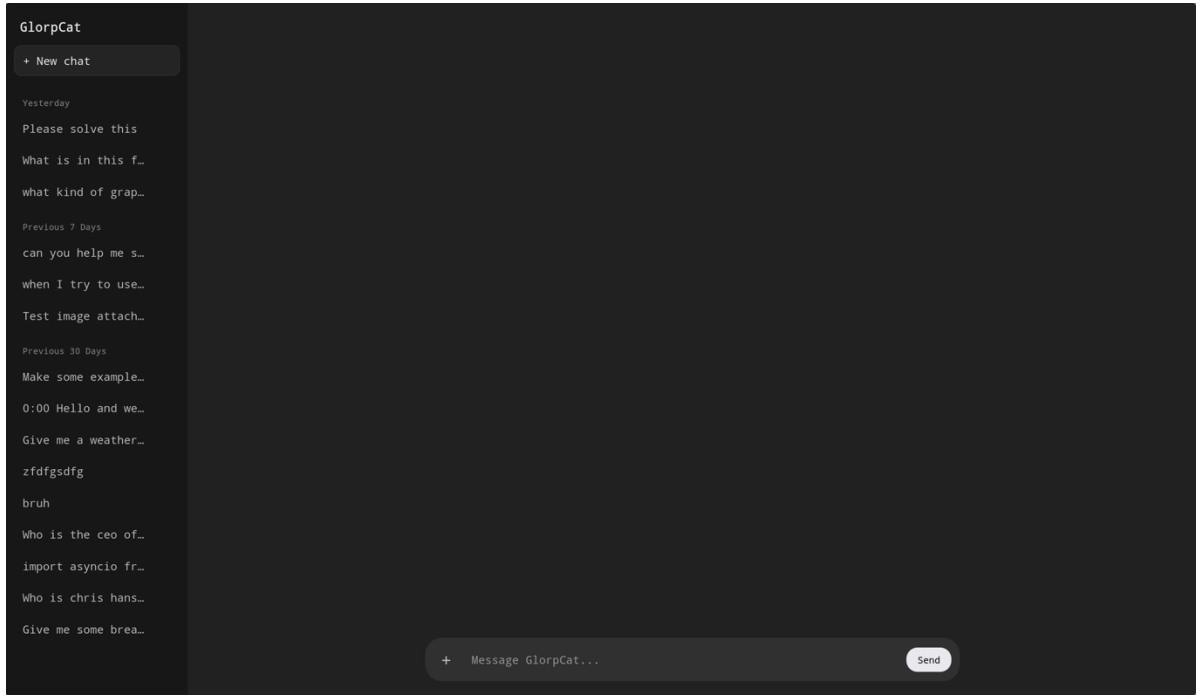
- A normal database stores documents
- Ours stores documents and vectors
- The machines read the vectors
  - Vectors find the relevant passage
- We read the documents
  - Original text, source, and page are stored verbatim

Vector machines read	**Text chunk** we read	**Source** we read
`[0.12, −0.83, 0.45, …]`	“A Digital Twin (DT) is a comprehensive digital replica of a physical asset…”	Sado et al. 2025, p. 1
`[−0.31, 0.08, 0.92, …]`	“Docling … runs efficiently on commodity hardware in a small resource budget.”	Auer et al. 2024, p. 1
`[0.55, 0.17, −0.40, …]`	“The periodic solution … is globally asymptotically stable from Lemma 4.”	Gao et al. 2026, near Eq. (7)

Each record keeps its vector, its original text, and where it came from. Nothing is dissolved into the model.

Agent

- Current Model: Nemotron 3.5 Lightning
  - 30 billion parameter mixture of experts model (MoE)
  - Model hosted through Ollama via Python
  - Browser UI
  - Tool Calling:
    - get_collections
    - query_collection
    - get_time
    - get_weather
    - search_text
    - search_images
    - search_news
    - search_videos
    - search_books
    - extract_content
    - sequentialthinking



Retrieval

- Retrieval of documents is based on tool use.
  - Get_collections
    - Checks what collections exist
  - Query_collection
    - Query collection
- Example document: arts_lab_graduate_handbook


ARTS-Lab Graduate Student Handbook

Austin R.J. Downey

July 27, 2026

1

Downey, Austin R. J. *ARTS-Lab Graduate Student Handbook*. 27 July 2026.

Overall Results

Performance:

Total Prompt Tokens (tokens):	144285	Average Prompt Speed (tok/s):	3111.16	Average Query Time (s):	0.18
Total Generated Tokens (tokens):	16766	Average Generation Speed (tok/s):	49.97	Average Response Time (s):	26.62
Total Tokens:	161051	Average Generated Tokens (tokens):	838	Average Prompt Tokens (tokens):	7,212

Questions:

1	What do I need to graduate with a Ph.D.?	Pass	11	Compare M.S. vs. Ph.D. expectations	Pass
2	What IDE does the lab recommend?	Pass	12	Summarize the software stack	Pass
3	Can I work remotely?	Pass	13	Can I graduate if my funding runs out?	Pass
4	Timeline for completing a Ph.D.	Fail	14	How should I spend my time at conferences?	Pass
5	How should I structure my dissertation?	Fail	15	Why do I need a “making_figures” folder?	Pass
6	What to do with figures in “making_figures”?	Pass	16	Font sizes in figures?	Pass
7	Do I have to say hi to Dr. Downey at a conference?	Fail	17	Who do we work for?	Pass
8	Expected version of M.S. thesis	Pass	18	What values does ARTS-Lab expect?	Pass
9	Compare qualifier, proposal, final defense	Pass	19	What is LaTeX?	Pass
10	How does cosine similarity work?	Pass	20	What is peer review?	Pass

Retrieval Versus Fine-Tuning

- Fine-tuning retrains the model on our documents
  - Knowledge dissolves into the weights
- Retrieval hands a frozen model an open book
  - Documents stay readable, outside the model
- Only retrieval can show what it used

	Fine-tuning	Retrieval (ours)
Knowledge lives in	Model weights	Readable documents
The model	Changed by training	Frozen, never changed
Cites its sources	No	Yes, every claim
Add a document	Retrain and revalidate	Add chunks, new snapshot
Remove a document	No clean way	Delete its chunks
Audit in 20 years	Inspect billions of numbers	Read the source text

Fine-tuning still helps with domain vocabulary and output style. The facts come from retrieval.

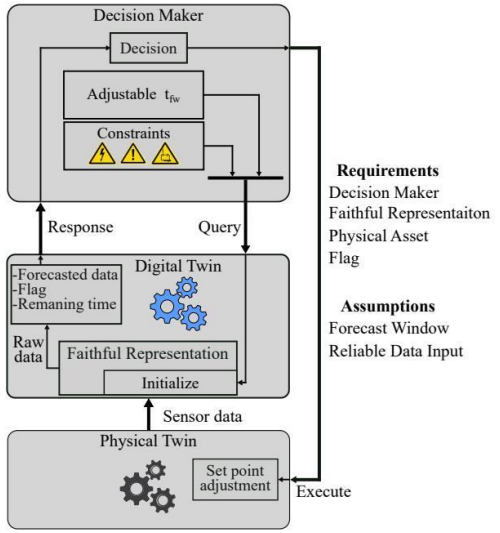
Scientific Merit and Publication

- Contribution: a retrieval agent designed for high trust
- Faithful scientific ingestion
  - Papers to Marker, other documents to Docling
- Provenance as the design target
  - Every claim traces to a stored chunk
- Frozen, offline, and edge-deployable
- A trusted assistant for deployable digital twins

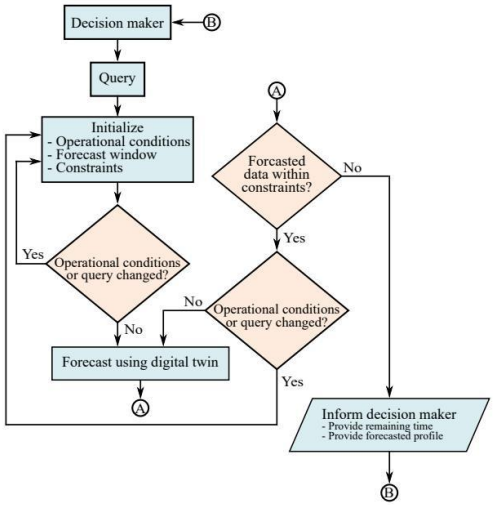


Other Applications

- Due to tool use, agent can be used to control systems
  - Such as digital twins
  - Knowledge-based decision making
  - Runs in an agent loop

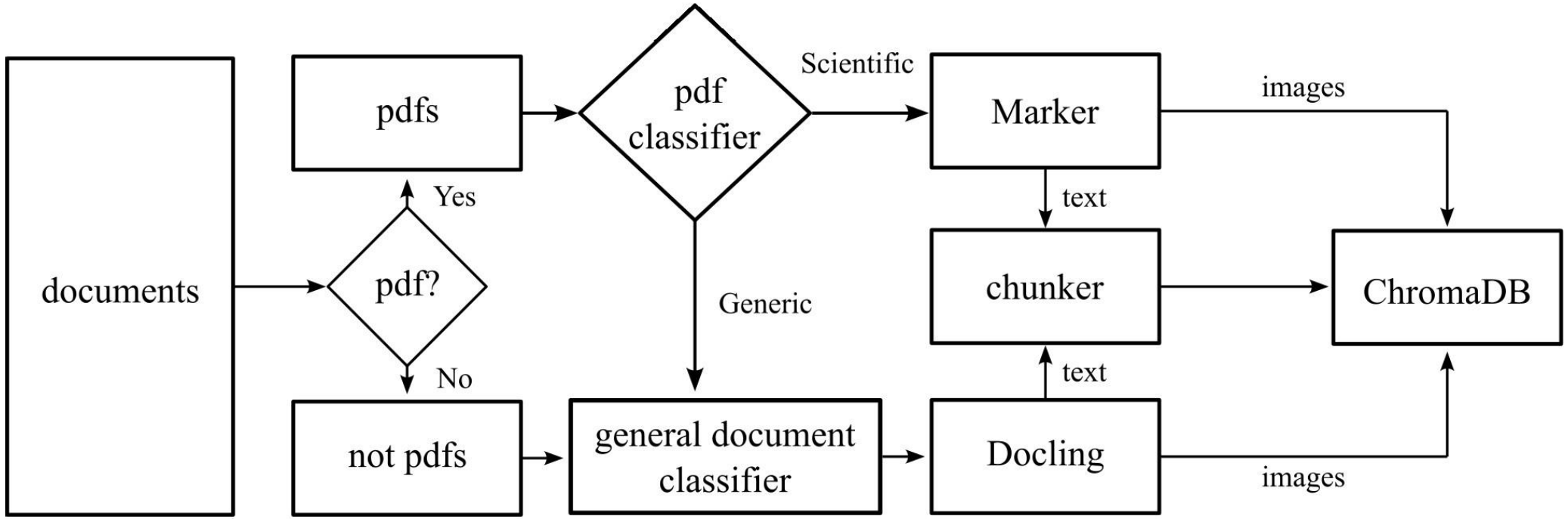


(a) Cyber-physical system.



(b) Forecasting routine for the digital twin.

Pipeline


documents
pdfs
pdf?
Yes
No
not pdfs
pdf classifier
Scientific
Generic
general document classifier
Marker
Docling
text
text
chunker
images
images
ChromaDB